

KANCHANAMALA P

malakanchana26@gmail.com 7299668990 CHENNAI Kanchanamala P kanchanamala-dev Portfolio Website

PROFILE

Second-year Electronics and Communication Engineering student with strong interest in problem-solving, embedded systems, and software development. Motivated to learn industry-level engineering practices and contribute through hands-on projects combining hardware and software.

EXPERIENCE

- 05/2026 – 07/2026
Chennai
- R&D Intern, Jedlik Motor Pvt. Ltd.**
 - Worked on STM32-based embedded motor control systems.
 - Performed motor profiling, firmware deployment, and system validation for BLDC/PMSM drives.
 - Worked on FOC based design of BLDC/PMSM motor drivers and control electronics.

EDUCATION

- 2024 – 2028
CHENNAI
- B.Tech in Electronics and Communication Engineering, VELLORE INSTITUTE OF TECHNOLOGY**
CGPA: 9.32 / 10
- 2023 – 2024
AVADI
- Senior Secondary, JGVV (CBSE BOARD)**
Score: 458 / 500 (91.6%)
- 2021 – 2022
AVADI
- Secondary, JGVV (CBSE BOARD)**
Score: 473 / 500 (94.6%)

PROJECTS

- 12/2025 – 04/2026
- Whistle-Based Smart Gas Stove Automation, Embedded Systems Project**
 - Developed an STM32-based system to detect cooker whistles and count user-defined targets.
 - Automated gas knob control using motor actuation for kitchen safety.
- 05/2025 – 04/2026
- Smart Home Energy Monitoring & Prediction Dashboard, IoT & Machine Learning Project**
 - Built a real-time energy monitoring system using Raspberry Pi, NodeMCU, InfluxDB, and Grafana.
 - Implemented a Python ML model to forecast household energy consumption.
- 07/2025 – 12/2025
- Morse Code Translator Using FPGA, FPGA & Digital Design Project**
 - Designed a Verilog-based FPGA system to convert Morse button inputs into ASCII characters.
 - Implemented timing logic for accurate symbol differentiation and decoding.
- 07/2025 – 12/2025
- Cell Phone Detector Using Analog Components** , Analog Circuits & PCB Project
 - Built a mobile signal detection circuit using antenna and LM357 op-amp.
 - Designed PCB and performed component placement, soldering, and debugging.
- 07/2024 – 12/2024
- IoT-Based Plant Monitoring and Watering System** , Embedded IoT Automation Project
 - Developed automated irrigation using NodeMCU and environmental sensors.
 - Enabled remote monitoring via mobile dashboard.
- 07/2024 – 12/2024
- Currency Detector and Total Amount Calculator** , Embedded Systems & Sensors Project
 - Designed IR/color sensor-based Indian currency detection system.
 - Implemented logic for note classification and total amount calculation.
- 12/2025 – 04/2026
- Junctionless Double Gate MOSFET Design & CMOS Analysis, TCAD & VLSI Simulation Project**
 - Simulating JLDG NMOS/PMOS in Synopsys Sentaurus and analyzing device characteristics.
 - Preparing matched devices for CMOS inverter implementation in Cadence Virtuoso.

TECHNICAL SKILLS

JAVA | Python | Embedded C | Verilog HDL | PCB Design (EasyEDA) | Assembly Programming | GRAFANA | MATLAB

INTERESTS

Internet of Things (IoT) | Embedded Systems | Automation and Control Systems | Analog and Mixed Signal Circuits

LEADERSHIP AND RESPONSIBILITIES

- 08/2024 – 08/2025
VIT CHENNAI
- Programme Representative – Electronics and Communication Engineering (ECE)**

Represented ECE students in academic and departmental matters, strengthening leadership, responsibility, and teamwork through addressing concerns and collaborating with faculty and peers.

CERTIFICATIONS & WORKSHOPS

- IoT Automation with Raspberry Pi – IIT Madras** [↗](#)
 - Participated in a hands-on workshop covering **NodeMCU, Raspberry Pi basics, and IoT automation concepts.**
- Unlocking the IoT Universe: Hands-on Masterclass – VIT Chennai** [↗](#)
 - Gained practical exposure to **NodeMCU interfacing and core IoT workflows** through guided sessions.

LANGUAGES

ENGLISH - Proficient | TAMIL - Native | FRENCH - Basic